

P09 Hq Solution

Origin of Matter-Antimatter Asymmetry (HQ v54)

Domain: Particle Physics/Cosmology

Model: PAX L5 Narrow L2 General 27B

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Run: PAX_27B HQ v54 Tokens: 7399 Time: 1292.1s Chain:

7c2a10f275308257b95335f6839c1565159650e399175675822ed873083c14f7 Genesis:

8b4a8a4f6312dfbe885de8280716985637c163fd2a4b5590341d56db1cc4e560

Subject: Proposed Resolution to the Baryon Asymmetry Problem via Resonant Leptogenesis with CP-Violating Neutrino Oscillations in the Early Universe

1. Core Hypothesis

The observed baryon asymmetry of the universe (BAU), quantified by $\eta_B \approx 6 \times 10^{-10}$, originates from **Resonant Leptogenesis** driven by the decay of heavy right-handed Majorana neutrinos (N_i) in the early universe. Unlike standard thermal leptogenesis which requires extremely high mass scales ($M_N > 10^9$ GeV) and often conflicts with low-energy neutrino data, this hypothesis posits that the necessary CP violation arises from **quasi-degenerate masses** of the lightest two right-handed neutrinos ($M_1 \approx M_2$). This degeneracy enhances the CP-asymmetry parameter ϵ_{CP} through resonant self-energy corrections, allowing successful baryogenesis at accessible energy scales ($M_N \sim 10$ TeV), potentially within reach of future colliders or precision intensity frontier experiments. The generated lepton asymmetry is subsequently converted into baryon asymmetry via non-perturbative electroweak sphaleron processes before they freeze out.

2. Mathematical Framework

The framework relies on the extension of the Standard Model (SM) by three right-handed singlet fermions N_R , leading to the Type-I Seesaw mechanism.

A. Lagrangian Extension:

$$\mathcal{L}_{\text{BSM}} = \mathcal{L}_{\text{SM}} - \frac{1}{2} \overline{N_R^c} M_R N_R - \left(\overline{L_L} Y_\nu \tilde{\Phi} N_R + \text{h.c.} \right)$$

Where:

* L_L : SM lepton doublet.

* Φ : Higgs doublet, $\tilde{\Phi} = i\sigma_2 \Phi^*$.

* Y_ν : Yukawa coupling matrix.

* M_R : Majorana mass matrix for N_R .

After Electroweak Symmetry Breaking (EWSB), the light neutrino mass matrix is given by the seesaw formula:

$$m_\nu \approx -v^2 Y_\nu M_R^{-1} Y_\nu^T$$

B. Resonant Enhancement of CP Asymmetry:

In the limit where $M_1 - M_2 \lesssim \Gamma_{1,2}/2$ (where Γ is the decay width), the CP asymmetry parameter ϵ_1 for the decay $N_1 \rightarrow \ell \phi$ is enhanced beyond the standard perturbative bound. The total asymmetry includes vertex and self-energy contributions:

$$\epsilon_1 = \frac{1}{8\pi} (Y_\nu^\dagger Y_\nu)_{11} \sum_{j \neq 1} \text{Im} \left[(Y_\nu^\dagger Y_\nu)_{1j}^2 \right] f(M_j^2/M_1^2)$$

For quasi-degenerate states ($M_1 \approx M_2$), the loop function $f(x)$ becomes dominated by the Breit-Wigner resonance:

$$\epsilon_1 \approx \frac{1}{8\pi} \frac{(Y_\nu^\dagger Y_\nu)_{12} \text{Im}(Y_\nu^\dagger Y_\nu)_{22}}{(Y_\nu^\dagger Y_\nu)_{12}^2} \frac{\Gamma_2}{(M_1 - M_2)^2 + (\Gamma_2/2)^2}$$

This allows ϵ_1 to approach $\mathcal{O}(1)$ even if Yukawa couplings are small, satisfying the Sakharov conditions efficiently.

C. Boltzmann Evolution:

The evolution of the number densities $n_{\{N_i\}}$ and the lepton asymmetry $n_{\{L-\alpha\}}$ is governed by coupled Boltzmann equations including washout effects:

$$\frac{dn_{\{N_i\}}}{dz} = -D_i(z) \left(\frac{n_{\{N_i\}}}{n_{\{N_i\}}^{eq}} - 1 \right)$$

$$\frac{dn_{\{L-\alpha\}}}{dz} = -\sum_i \epsilon_{\{i,\alpha\}} D_i(z) \left(\frac{n_{\{N_i\}}}{n_{\{N_i\}}^{eq}} - 1 \right) - W_{\alpha}(z) n_{\{L-\alpha\}}$$

Where $z = T/M_1$, D_i represents the decay rate, and W_{α} represents the inverse decay and scattering washout terms.

3. Proof Sketch / Key Steps

1. Sakharov Condition Verification:

- * **Baryon Number Violation**: Provided by non-perturbative sphalerons in the SM plasma above $T_c \approx 130$ GeV.

- * **C and CP Violation**: Generated by complex phases in the Yukawa matrix Y_{ν} during N_i decays, amplified by the resonant denominator $(M_1 - M_2)^2 + \Gamma^2/4$.

- * **Departure from Thermal Equilibrium**: Achieved when the decay rate $\Gamma_{N_1} < H(T=M_1)$ (Hubble expansion rate), ensuring the process occurs out-of-equilibrium.

2. Derivation of the Resonance Condition:

- * Demonstrate that for hierarchical masses, $\epsilon_1 \propto M_1$. To match η_B , one needs $M_1 > 10^9$ GeV.

- * Show mathematically that introducing a splitting $\Delta M = M_1 - M_2$ such that $\Delta M \sim \Gamma$ creates a pole in the propagator correction, boosting ϵ_1 by a factor of $\sim M/\Gamma$.

- * Calculate the required fine-tuning: $\Delta M / M \sim 10^{-8}$ to 10^{-12} depending on the specific Yukawa texture, arguing this is natural in models with approximate flavor symmetries (e.g., $SU(2)_R$).

3. Flavor Effects Integration:

- * Incorporate "flavored" Boltzmann equations. At temperatures $T < 10^9$ GeV, charged lepton Yukawa interactions maintain coherence between flavors.

- * Prove that flavor correlations can suppress washout rates for specific lepton flavors while preserving asymmetry in others, significantly relaxing the constraints on M_N .

4. Conversion to Baryon Asymmetry:

- * Apply the sphaleron conversion factor: $n_B = c_{\text{sph}} n_L$. In the SM, $c_{\text{sph}} \approx -28/79$.

- * Integrate the final lepton asymmetry Y_L over the temperature history to yield the final baryon-to-photon ratio $\eta_B = n_B/n_{\gamma}$.

4. Testable Predictions

If this hypothesis holds, it yields distinct experimental signatures distinguishable from high-scale leptogenesis:

1. Low-Scale Heavy Neutral Leptons (HNLs):

- * Existence of sterile neutrinos with masses in the MeV–TeV range.

- * **Search Strategy**: Displaced vertices at LHCb, ATLAS, CMS, or dedicated experiments like SHiP, FASER, and DUNE near detectors.

2. Lepton Flavor Universality Violation (LFUV):

- * Specific patterns in rare meson decays (e.g., $K \rightarrow \pi \nu \bar{\nu}$, $B \rightarrow K \mu e$) induced by mixing between active and sterile neutrinos.

3. Neutrinoless Double Beta Decay ($0\nu\beta\beta$):

- * While standard seesaw predicts very long half-lives due to tiny m_{ν} , the low-scale resonant scenario may allow observable rates if the effective Majorana mass m_{ee} receives contributions from the heavy sector interference or if the light neutrino spectrum is inverted/quasi-degenerate.

4. Cosmic Microwave Background (CMB) Constraints:

- * Precise measurements of the effective number of relativistic species (N_{eff}) and primordial helium abundance (Y_p) could constrain the lifetime and decay modes of these HNLs in the early universe.

5. Gravitational Waves:

* If the phase transition associated with the generation of asymmetry involves a strong first-order phase transition (linked to the symmetry breaking scale), it would generate a stochastic gravitational wave background detectable by LISA or DECIGO.

5. Remaining Open Sub-Problems

Despite the theoretical elegance, several critical challenges remain:

1. **Fine-Tuning vs. Naturalness:**

* Achieving the precise mass degeneracy ($\Delta M \sim \Gamma$) required for resonance enhancement without an underlying protective symmetry remains difficult. Is there a dynamical mechanism (e.g., radiative stability) that enforces this degeneracy?

2. **Thermal Corrections and Plasma Effects:**

* Accurate calculation of the thermal widths $\Gamma(T)$ and thermal masses in the dense early universe plasma is computationally intensive. Current approximations may underestimate washout effects or shift the resonance condition.

3. **Connection to Low-Energy Neutrino Data:**

* Mapping the high-energy parameters (Y_ν, M_R) to the measured low-energy observables (m_ν , PMNS mixing angles, Dirac CP phase δ_{CP}) is highly non-trivial. There is a vast parameter space; determining if the specific textures required for resonant leptogenesis are compatible with current oscillation data (especially the normal vs. inverted hierarchy preference) is ongoing.

4. **Non-Perturbative Sphaleron Dynamics:**

* The exact efficiency of the sphaleron conversion depends on the detailed dynamics of the electroweak phase transition. If the transition is a smooth crossover (as suggested by lattice QCD for the SM), the window for converting lepton asymmetry to baryon asymmetry narrows, requiring extensions to the scalar sector (e.g., Two-Higgs-Doublet Models) to ensure a strong first-order phase transition.

5. **Initial Conditions:**

* The solution assumes a thermal initial state. However, inflationary reheating scenarios might produce non-thermal distributions of N_i , altering the Boltzmann evolution entirely. A robust theory must be independent of specific reheating histories.